

Hyperbaric Vacuum-Based Artificial Muscles for High-Performance Actuation

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Existing pneumatic artificial muscles generally rely on the use of either positive or negative pressure for actuation with each having their own advantages and disadvantages. Herein, a hyperbaric vacuum-based artificial muscle (Hyper-VAM) making use of both positive and negative pressures almost interchangeably is described. This is done by placing a vacuum-based actuator inside a hyperbaric chamber which enables both the large deformations possible using negative pressure, and large forces through the action of high pressure differential, exceeding those possible using only negative pressure. This actuator can lift heavy payloads (up to 80 kg) for its entire contraction range, with large deformations (89.1% contraction of the active length) and high power (maximum power of 101.3 W), while using relatively low pressures, which can be supplied by a portable pneumatic pump. By exchanging air between the two chambers in closed-loop pneumatic actuation, this actuator can also be driven using a single pump without exchanging air with the environment and its position can be controlled using a single pneumatic regulator.

Many pneumatic artificial muscles (PAMs) work through the pressurization of a deformable volume to produce a linear contraction through the lateral expansion of the volume. Cylindrical structures composed of rubbers or polymers reinforced with fibers called PAMs or McKibben muscles have been proposed using this principle.^[3–5] These actuators have been made in thinner form factors,^[6,7] have used expanding origami structures and unfolding pleats,^[8–10] have taken torsional pasta-inspired shapes,^[11] and have been made from thin airtight fabrics which make them lighter and easier to manufacture.^[12–16] However, the actuation performance of pressurized artificial muscles producing a contractile force is limited due to the constraints generated by the dichotomy of producing a contraction of the actuator through an expansion in volume.

1. Introduction

Soft pneumatic robots capable of large and adaptive deformations hold significant promise for building human-friendly robots for domestic and industrial use.^[1,2] The use of air makes them inherently compliant and significantly lighter than conventional robots, but current artificial muscles do not have sufficient performance to replace the electric motors used in today's industrial and cooperative robots for most real-world applications. The development of stronger, faster, more efficient, and lighter linear artificial muscles that are less dependent on bulky pneumatic sources will lead to robots making use of soft robotic technology that can be used for a wider range of applications.

Although the pressurization of a volume has so far been the predominant form of pneumatic actuation, it is also possible to depressurize this volume to create a negative pressure differential with the environment for soft actuation. Vacuum pressure has been used to produce the linear contraction of polymeric actuators that have capabilities resembling those of human muscles.^[17–19] Similar actuators have been 3D printed,^[20,21] and film-based vacuum actuators with reinforcements are capable of large and sustained forces with large displacements.^[22–26] It is possible to use vacuum pressure to produce other types of actuators such as bending actuators,^[20,23] joints with two degrees of freedom,^[27,28] twisting actuators,^[29] and origami actuators that can be used as grippers.^[30] However, the performance of vacuum-based artificial muscles is limited by the maximum pressure differential of 101.3 kPa that can be reached between ambient pressure and a perfect vacuum.

Some recent works have blurred the lines between negative and positive pressure artificial muscles. A linear actuator which is pressurized to extend and contracts through depressurization toward atmospheric pressure was shown to be able to produce contraction ratios up to 75%,^[31] and an actuator was demonstrated where the deformable volume is maintained at room pressure and placed within a tube which is itself pressurized.^[32] Linear and bending actuators making use of positive pressure chambers and negative pressure chambers have been demonstrated,^[33,34] but the use of the positive pressure reduces the force of the actuator throughout the later part

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of the deformation due the positive and negative pressure chambers having different maximum equilibrium deformations.

In this article, we present an artificial muscle consisting of a vacuum-based soft actuator placed within a hyperbaric chamber. This allows the actuator to make use of positive and negative pressures nearly interchangeably as both pressures act on the same deformable surfaces with the same maximum deformation. This concept can be used to generate a negative pressure differential across the surface of the actuator which is not limited by the atmospheric pressure and gives an additional degree of freedom to control its actuation behavior. The proposed hyperbaric vacuum-based artificial muscle (Hyper-VAM) can lift heavy payloads (up to 80 kg) for its full actuation range (89.1% contraction ratio of the active length) with high power (maximum power of 101.3 W). It can also be operated by exchanging air back and forth from the negative pressure chamber to the positive pressure chamber without exchanging air with the environment. This article introduces the design and manufacturing of the actuator, followed by the evaluation of its actuation performance, and the implementation of the control of the actuator in closed-loop pneumatic operation.

2. Actuator Concept

2.1. Hyper-VAM Design

The proposed actuator consists of a vacuum-based soft actuator combining a flexible membrane with rigid reinforcements which is placed within a hyperbaric chamber. The vacuum-based actuator used in this work is based on the authors' previous work and consists of a sealed film chamber connected to rigid top and bottom plates in which rigid lateral reinforcements are placed throughout the length at regular intervals (**Figure 1a**).^[22] Some minor modifications were made to this design such as changing the shape of the reinforcements into an octagonal shape and adding a long shaft protruding from the bottom plate. The actuator design allows for either positive or negative pressures to be used to create a negative pressure differential across the membrane of the actuator (**Figure 1b**), and the use of simultaneous positive and negative pressures results in an even greater negative pressure differential. This actuation principle poses additional pneumatic requirements due to having to simultaneously pull air from one volume and add air to another volume, which gives two degrees of freedom in the control of the actuator.

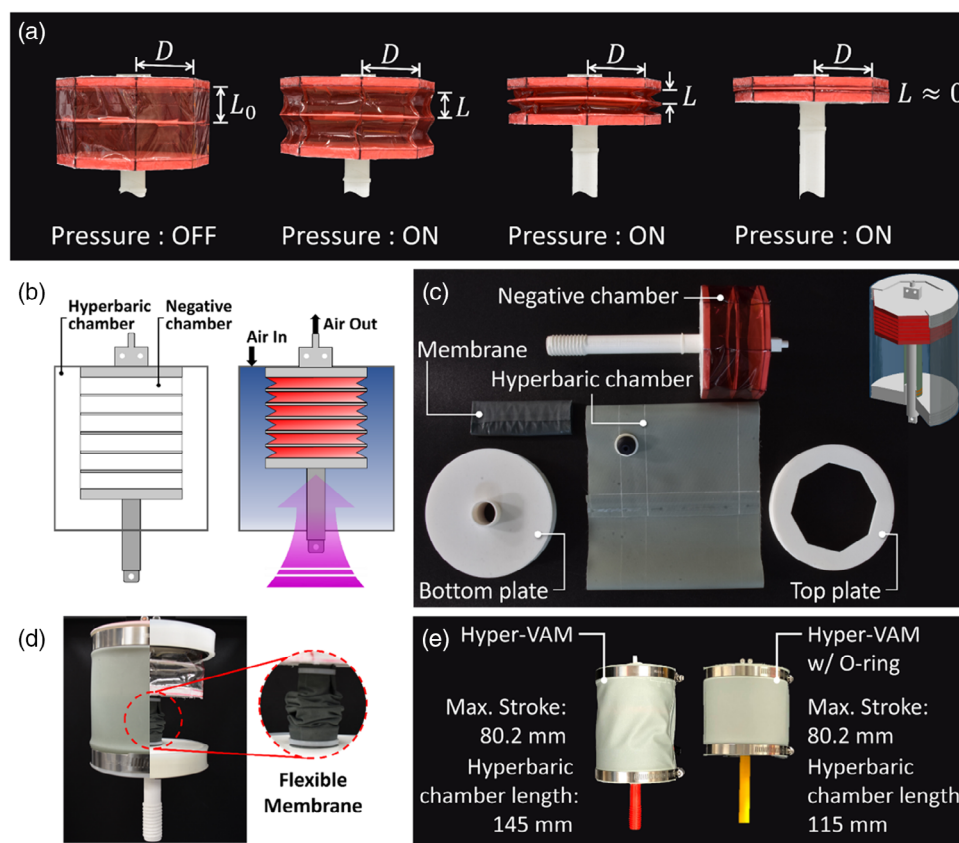


Figure 1. An overview of the concept of the Hyper-VAM. a) The vacuum-based actuator which is placed within the hyperbaric chamber and is capable of large contractions. b) Concept of the actuator showing how it can make use of both negative and positive pressures to create a negative pressure differential. c) Assembly of the Hyper-VAM actuator with a soft hyperbaric chamber. d) Membrane connected to the bottom of both chambers, which maintains the hyperbaric chamber hermetic throughout the motion. e) Fabricated Hyper-VAM actuators where the one on the left uses a flexible membrane for sealing of the positive chamber and the one on the right an O-ring.

One of the requirements of the proposed Hyper-VAM is that the top of the hyperbaric chamber and of the vacuum actuator must be fixed together while the bottom plate of the negative pressure chamber can independently move (Figure 1c). The base plate of the vacuum actuator has a long protruding shaft onto which the payloads are attached. This shaft can slide through a hole in the hyperbaric chamber while an airtight sleeve covers this shaft, sleeve that is connected to each bottom plate. This allows this sleeve cover to roll up and down throughout the motion of the shaft while keeping the hyperbaric chamber airtight (Figure 1d). This flexible sleeve can also be replaced by an O-ring which seals the shaft around the positive pressure chamber (Figure 1e). This paper will focus on the former, but the operation of the later type will also be demonstrated. Additional details on the manufacturing steps of the actuator, on the materials used, and on the dimensions of the actuators used in each experiment are included in the Experimental Section.

2.2. Operational Principle

Assuming that the membranes of the vacuum actuator easily deform without stretching and that no significant temperature changes occur in either chamber, the work of the fluid being added into the hyperbaric chamber and of the fluid removed from the vacuum chamber are both transformed into mechanical work. The principle of energy conservation can be used to determine the virtual translation dL corresponding to a change in volume V_{pos} in the positive pressure hyperbaric chamber and V_{neg} in the negative pressure vacuum chamber, that is

$$FdL - W_{\text{other}} = P_{\text{pos}}dV_{\text{pos}} + P_{\text{neg}}dV_{\text{neg}} \quad (1)$$

where F is the tension force, W_{other} is the work done by other forces such as the friction of the sleeve on the shaft, P_{pos} is the inner pressure of the hyperbaric chamber, P_{neg} is the inner pressure of the vacuum chamber, dV_{pos} is the derivative of the volume of the hyperbaric chamber, and dV_{neg} is the derivative of the volume of the hyperbaric chamber. Both pressures are gauge pressures such that P_{pos} is a positive value and P_{neg} is a negative value. As dV_{neg} is also negative during a contraction, its multiplication with the gauge pressure will yield a positive number which will be summed with the positive work generated by the positive pressure volume. It is to be noted that a small change in volume will occur in the outer layer of the hyperbaric chamber as the chamber expands slightly, but this will not produce any work and is not considered in the model.

The change in volume of the negative chamber produces an opposite change in volume of the hyperbaric chamber minus the volume of the shaft. Assuming that no frictional forces or forces are required to deform the structure of the actuator, Equation (1) can be simplified as follows.

$$F = (P_{\text{pos}} - P_{\text{neg}}) \frac{dV_{\text{neg}}}{dL} - P_{\text{pos}}A_{\text{shaft}} \quad (2)$$

where A_{shaft} is the cross-sectional area of the shaft. The contraction of the negative pressure chamber can then be separated into a first portion where the membrane of the negative pressure chamber folds inward in a cylindrical shape at each edge in the early portion of the deformation and a second portion where the membrane starts sticking to the rigid elements in the later portion of the deformation (Figure 2a–d). The volume of the negative pressure chamber in the first portion of the motion can be written as

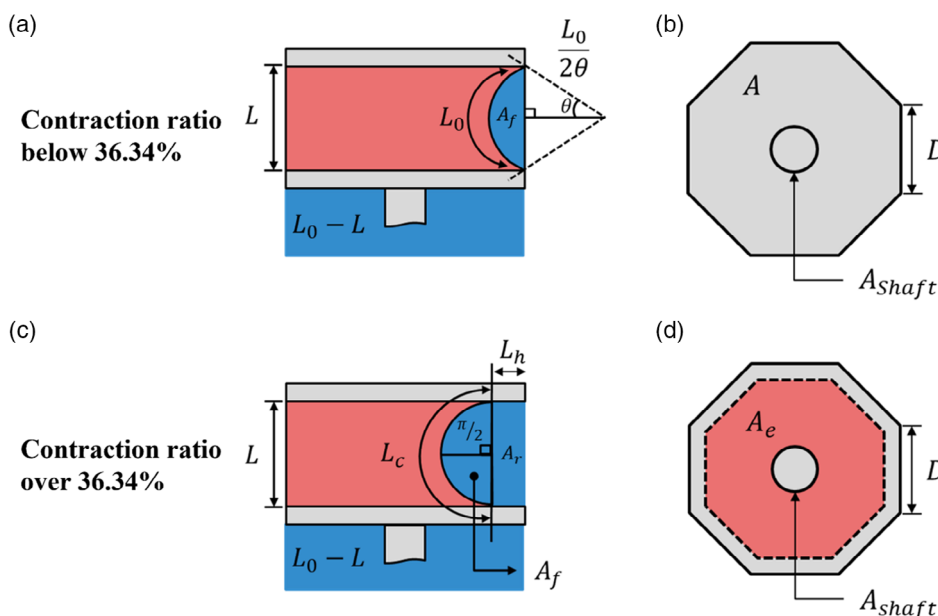


Figure 2. Geometry model and analysis of the change in volume within the Hyper-VAM. a) Side view of the change in volume before reaching a contraction ratio of 36.34%. b) Top view of the actuator bottom plate taking into consideration the area of the shaft and the octagonal area of the actuator. c) Side view of the actuator contraction ratio over 36.34%. d) Representation of the effective area once the actuator reaches 36.34% viewed from the top.

$$V_{\text{neg}} = \frac{2D^2 L_0 (1 + \sqrt{2}) \sin \theta}{\theta} - \frac{2DL_0^2}{\theta} \left(1 - \frac{\sin \theta \cos \theta}{\theta} \right) \quad (3)$$

where D is the side length of the top or bottom plate, L_0 is the initial distance between successive rigid parts before contraction, and θ is the inner angle of the circular cross section. The derivative with respect to the current length L and the total force F_{tot} in this portion of the deformation can then be obtained as

$$\frac{dV_{\text{neg}}}{dL} = \frac{\frac{dV_{\text{neg}}}{d\theta}}{\frac{dL}{d\theta}} = \left(2D^2 (1 + \sqrt{2}) + \frac{2DL_0}{\theta \cos \theta - \sin \theta} \left(1 + \frac{\theta \cos 2\theta - \sin 2\theta}{\theta} \right) \right) \quad (4)$$

$$F_{\text{tot}} = (P_{\text{pos}} - P_{\text{neg}}) \left(2D^2 (1 + \sqrt{2}) + \frac{2DL_0}{\theta \cos \theta - \sin \theta} \left(1 + \frac{\theta \cos 2\theta - \sin 2\theta}{\theta} \right) \right) - P_{\text{pos}} \pi r^2 \quad (5)$$

This force is valid up to a value of θ equal to 90° which is when the membrane starts to adhere to the rigid parts. This also corresponds to a contraction ratio of 36.34%. The volume in the later part of the deformation, its derivative with respect to the length, and the total force can be written as

$$V_{\text{neg}} = 2D^2 L (1 + \sqrt{2}) - (L^2 D \pi - 8LL_h D) \quad (6)$$

$$\begin{aligned} \frac{dV_{\text{neg}}}{dL} &= \frac{\frac{dV_{\text{neg}}}{d\theta}}{\frac{dL}{d\theta}} \\ &= -P_{\text{neg}} \left(2D^2 (1 + \sqrt{2}) + \frac{2DL_0}{\theta \cos \theta - \sin \theta} \left(1 + \frac{\theta \cos 2\theta - \sin 2\theta}{\theta} \right) \right) \end{aligned} \quad (7)$$

$$F_{\text{tot}} = (P_{\text{pos}} - P_{\text{neg}}) \left(2D^2 (1 + \sqrt{2}) + \frac{2DL_0}{\theta \cos \theta - \sin \theta} \left(1 + \frac{\theta \cos 2\theta - \sin 2\theta}{\theta} \right) \right) - P_{\text{pos}} \pi r^2 \quad (8)$$

Further information about the derivation of Equation (3) and (6) is included in the Supporting Information together with Equation S1–S18, Supporting Information.

3. Actuator Characterization

3.1. Blocked Force

The blocked force of the actuator with the flexible membrane was measured using positive pressure only and negative pressure only by fixing the top of the actuator and connecting the shaft of the actuator to a load cell. The linear contractile force of the actuator was measured at contraction ratios of the vacuum chamber of 25, 50, and 75% with a pressure ranging from 0 to 40 kPa (Figure 3a,b). The results show that the force is proportional to the pressure at all three contraction ratios and that the

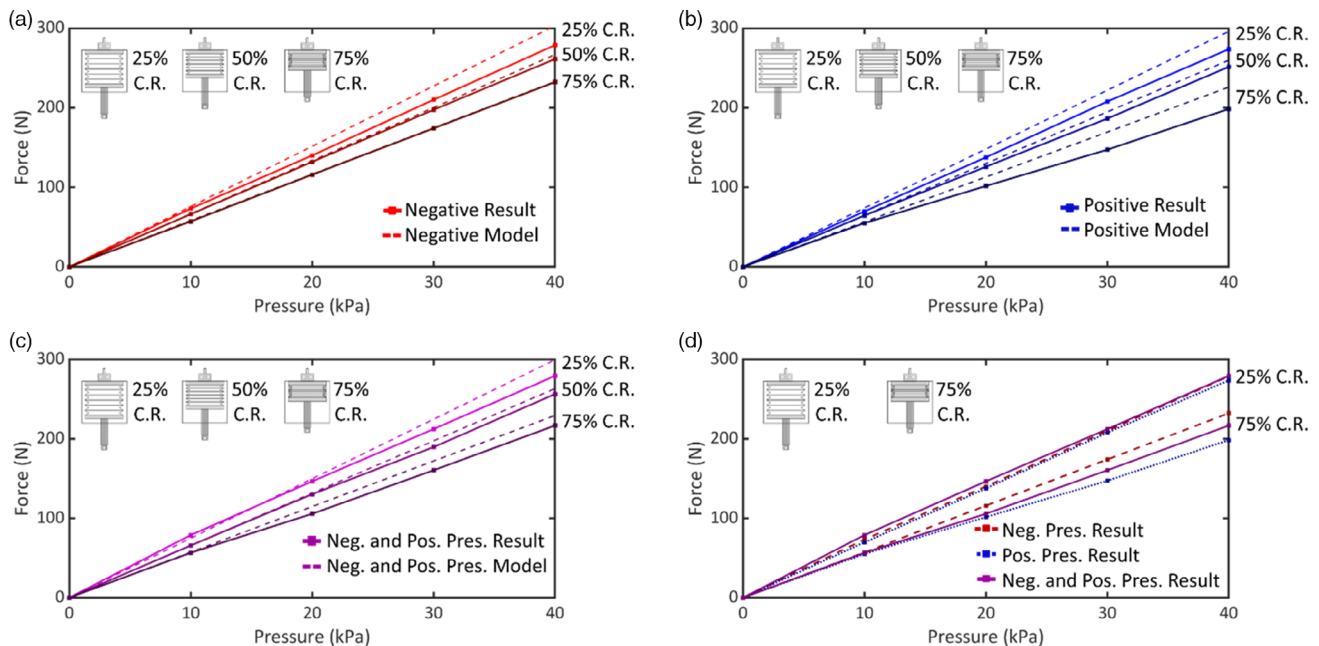


Figure 3. Actuation force results of the Hyper-VAM. a) Blocked force of the actuator using negative pressures for three different contraction ratios of the active length. b) Blocked force of the actuator using positive pressures for the same three different contraction ratios of the active length. c) Blocked force of the actuator using equal but opposite positive and negative pressures. d) Comparison of the blocked force of the actuator using negative pressure only, positive pressure only, and combined positive and negative pressures.

force produced by the actuator decreases slightly at higher contraction ratios. The force decrease is also more significant when using positive pressure than when using negative pressure but is still relatively steady compared with other PAMs.

The blocked force of the actuator was then measured for equal but opposite positive and negative pressures (Figure 3c). The linear contractile force of the actuator was measured at similar contraction ratios but with a total pressure differential ranging from 0 to 40 kPa such that the maximum tested pressure is -20 kPa in the negative pressure chamber and $+20$ kPa in the positive pressure chamber. The results show that the force is proportional to the pressure at all three contraction ratios with the force reaching 279.2 N at 40 kPa of pressure and that the force produced by the actuator decreases slightly at higher contraction ratios. The experimental results show a good fit with the modeling results.

Comparing these results shows that combined positive and negative pressures fall in between using only positive or negative pressures (Figure 3d). At higher contraction ratios and at the same total pressure differential, negative pressure actuation produces a higher blocked force than either positive or hypervacuum pressures while using only positive pressure yields the lowest blocked force. This is due to the smaller volume change of the hyperbaric chamber due to the volume occupied by the shaft. As the combined negative and positive pressure experiment was set to include only half of its pressure differential from positive pressure, the effect of the shaft on its blocked force is also halved.

3.2. Isobaric Force

The isobaric force of the actuator when using positive pressure only, negative pressure only, and equal but opposite pressures was measured using a linear tensile testing machine up to just before the maximum contraction of the actuator (Figure 4a–c). The force produced by the actuator decreases steadily throughout the contraction as is the case for most PAMs but maintains more than half of its maximum force up to its maximum contraction except when using positive pressure only where it drops to slightly below half of its maximum force. Comparing the performance of the actuator with hypervacuum pressure versus negative and positive pressures shows that negative pressure has the least reduction in force throughout the contraction and that positive pressure has the most reduction in force (Figure 4d). This difference between the pressures is likely due to the presence of the shaft in the positive pressure chamber and due to friction of the sleeve cover under positive pressure. These results show that the proposed Hyper-VAM can produce large forces using positive pressure, negative pressure, or combined pressures.

3.3. Contraction Ratio Tests

One of the particularities of having a negative pressure chamber within a positive pressure chamber is that the displacement is

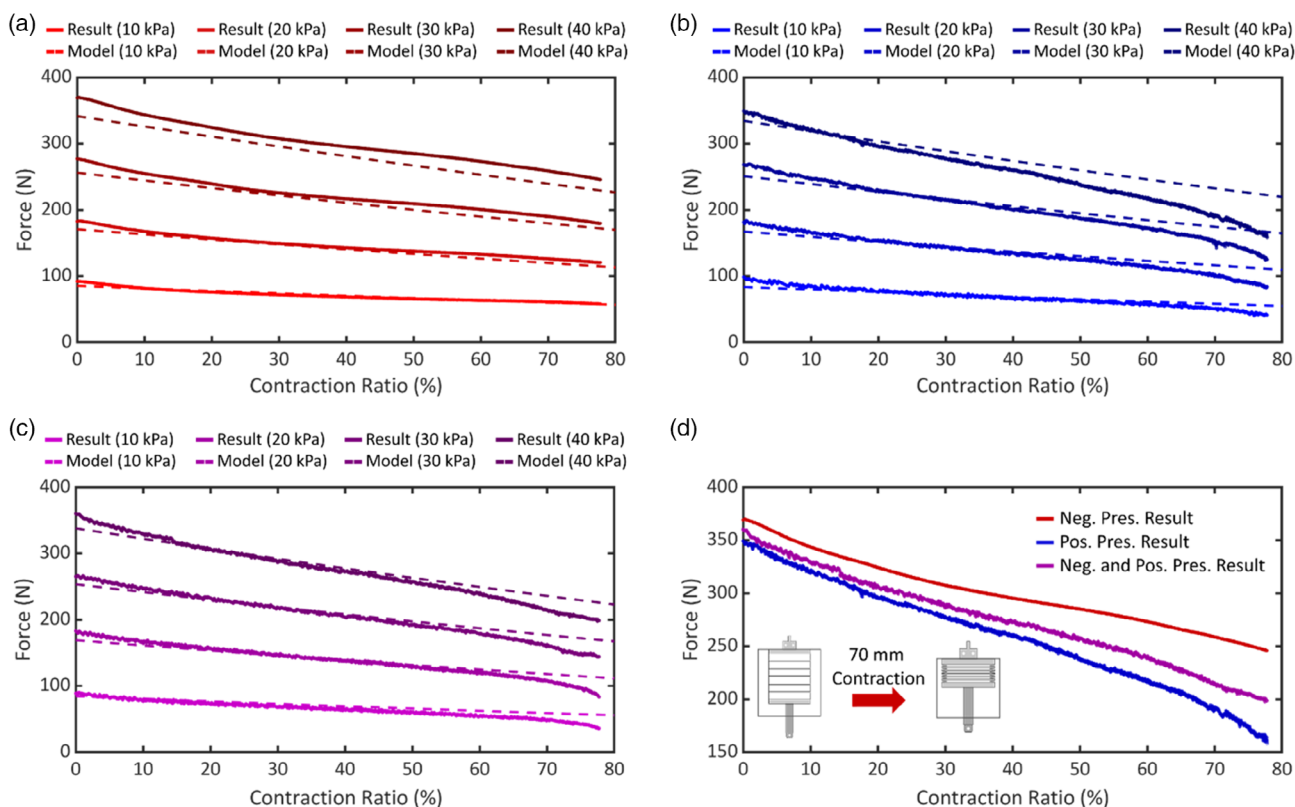


Figure 4. Isobaric force results of the Hyper-VAM while for a 70 mm stroke. a) Isobaric force of the actuator using negative pressures. b) Isobaric force of the actuator using equal but opposite positive pressures. c) Isobaric force of the actuator using equal but opposite positive and negative pressures. d) Comparison of the isobaric force of the actuator using negative pressure only, positive pressure only, and combined positive and negative pressures at 40 kPa.

done by the negative pressure chamber, but the length of the actuator is set by the length of the positive pressure chamber, which does not move significantly throughout its displacement. This means that the contraction ratio can be calculated relative to the length of the negative pressure chamber or the positive pressure chamber. Two different configurations of positive pressure chamber with the same negative pressure chamber will be tested where one uses a flexible membrane for sealing of the positive pressure around the shaft and the other an O-ring (Figure 1e). Both negative pressure chambers have an active length of 90 mm and the positive pressure chamber with the flexible membrane has an active length of 127 mm and with the O-ring an active length of 100 mm. The active length is defined as the distance between the top and bottom plate of each chamber rather than the total length which includes the thickness of these plates. This difference in inner length is due to the length of the folded membrane when the negative pressure chamber is extended.

Both actuators were loaded with a payload of 30 kg and actuated with equal but opposite pressures of 40 kPa (Figure 5a,b). Both actuators produced a full contraction of 80.2 mm, which corresponds to the full contraction of the negative pressure chamber and 89.1% of the active length of the negative pressure chamber. However, this corresponds to 63.1% of the active length of the positive pressure chamber for the actuator using a flexible membrane for sealing of the positive pressure chamber and 80.2% of the active length for the actuator using the O-ring. However, the O-ring is more likely to cause leaks and friction on the shaft than using the flexible membrane.

3.4. Maximum Payload Tests

One of the disadvantages of vacuum-based artificial muscles is that their maximum pressure differential is limited by the difference between a perfect vacuum and room pressure. The advantage of the present actuator is that it is not limited by this pressure differential as the positive pressure chamber effectively increases the pressure of the atmosphere around the negative pressure chamber. The actuation force limit of the Hyper-VAM was tested using a load of 80 kg and applying a negative pressure of 90 kPa and a positive pressure of 60 kPa for a total pressure differential of 150 kPa. The Hyper-VAM was able to produce a linear deformation of 80.2 mm with this payload (Figure 6a, Movie S1, Supporting Information). This

corresponds to its full contraction. The normalized average work per active length of the actuator during this demonstration is 674.93 J m^{-1} . As part of a robotic arm, this linear force is sufficient to be used as a bicep muscle that can produce large lifting deformations even with payloads of 5 kg at a moment arm of 30 cm (Figure 6b, Movie S2, Supporting Information).

3.5. Actuation Speed

The dynamic range of the actuator was tested with a payload of 10 kg by simultaneously pressurizing the positive pressure chamber and vacuuming the negative pressure chamber until it reaches a target upper deformation and then, by a solenoid valve, the flow is inverted until it reaches a target lower deformation (Figure 6c). This flow inversion results in both chambers returning toward their original pressures and allows the actuator to extend faster than through exchanging air with the environment. With a displacement range from 6 to 72 mm, this resulted in a cycling frequency of $\approx 0.5 \text{ Hz}$, from 20 to 60 mm of $\approx 0.75 \text{ Hz}$, and from 35 to 45 mm of $\approx 3.23 \text{ Hz}$ (Figure 6d, Movie S3, Supporting Information).

Next, the potential of the actuator for producing high-speed actuation and high power is demonstrated with a payload of 40 kg. The pressurized tank of a pneumatic pump was connected directly to the positive pressure chamber, a vacuum pump was connected directly to the negative pressure chamber, and both sources were simultaneously activated. The actuator produced a full contraction with the 40 kg payload in 0.31 s (Figure 6e,f, Movie S4, Supporting Information), which corresponds to an average power of 101.3 W. A small overshoot and oscillation at the end of the motion can be observed as the momentum of the payload is sufficient to cause some unintended deformations of the actuator and setup.

4. Portable Operation of Robotic Arm

4.1. Closed-Loop Pneumatic Actuation and Control

Previous experiments were conducted with nonportable air compressors or vacuum generators, but real operating conditions will often be restricted in the type of pump used and how this pump is used will affect the performance of the system. In some scenarios, it may not be possible or desirable to use large pneumatic

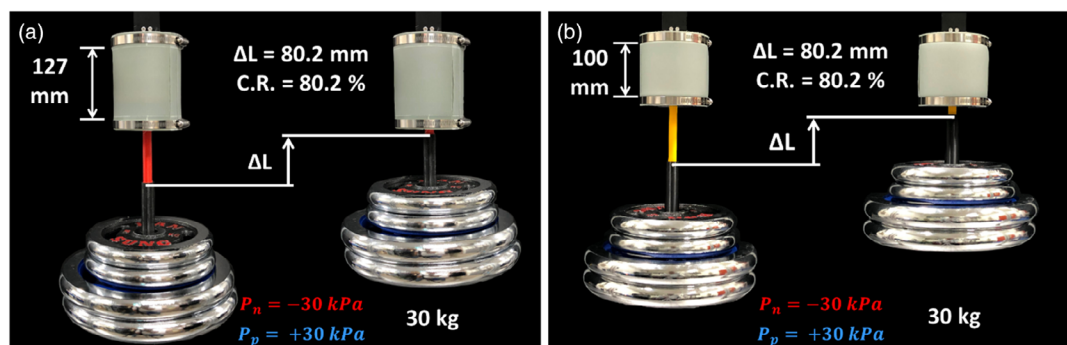


Figure 5. Contraction ratio of the active length comparison for both Hyper-VAMs. a) Hyper-VAM using the flexible material for sealing of the positive pressure within the actuator. b) Hyper-VAM with an O-ring built-in to airtight the hyperbaric pressure within the actuator.

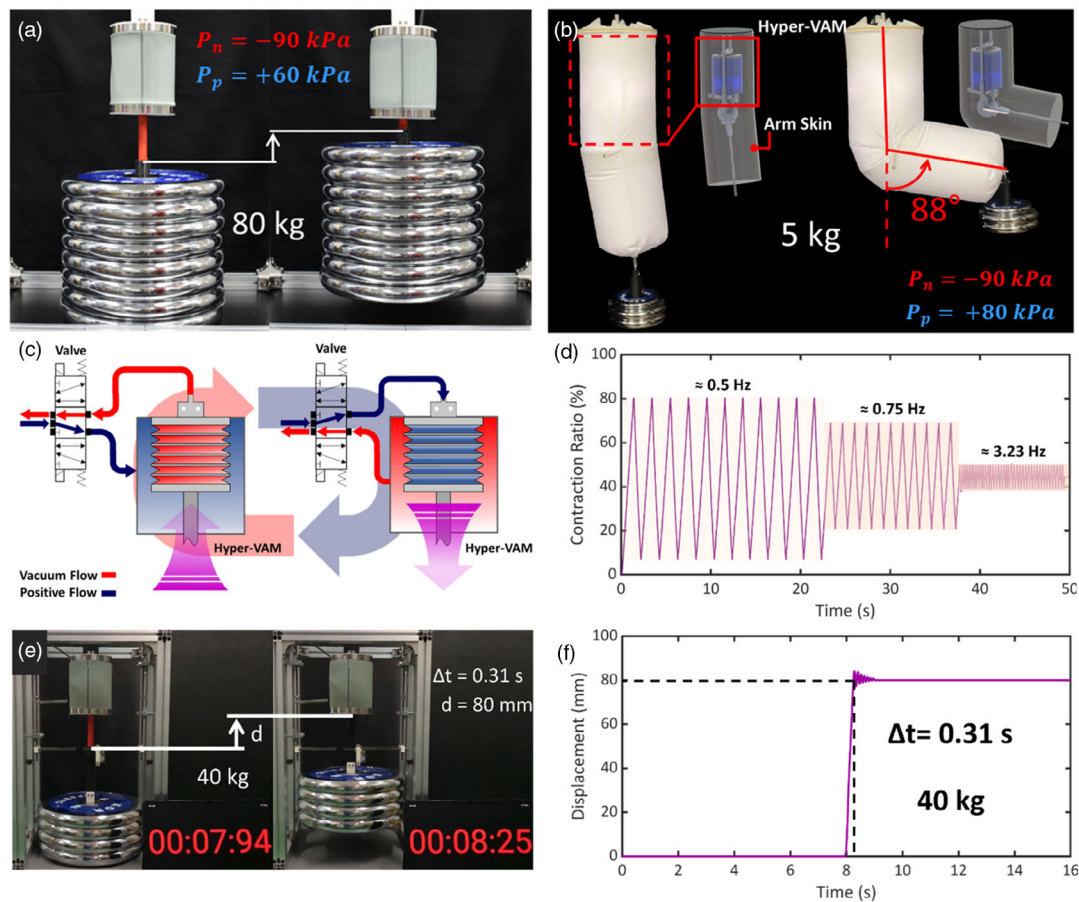


Figure 6. Payload and speed test results of the Hyper-VAM. a) The Hyper-VAM producing a full contraction with a payload of 80 kg. b) The Hyper-VAM implemented as the bicep of a robotic arm with an inflated skin lifting a payload of 5 kg with a moment arm of 30 cm. c) Test concept for repeated actuation and release for dynamic range testing. d) Dynamic range results for three different actuation ranges. e) Maximum power test with a payload of 40 kg. f) Displacement during the maximum power test.

devices. In particular, the proposed actuator uses two pneumatic chambers such that separate operation of both chambers would require two pumps and two pneumatic controllers. However, as one functions on positive pressure and the other on negative pressure, it is possible to pump air from the negative pressure chamber to the positive pressure chamber for actuation and to allow air to flow in the opposite direction for relaxation. This would allow for closed-loop pneumatic operation without exchanging air with the environment.

The Hyper-VAM using closed-loop pneumatic actuation and a portable pneumatic pump was implemented for driving a flexible gripper application (Figure 7a). The actuator was implemented into a 3D-printed gripper mechanism with a flexible material for the gripper's structure and a rigid material used for the parts used to secure the actuator in place. The gripper was tested for different items with different shapes, textures, and weights up to 3 kg. The mechanism was able to grasp a variety of objects, demonstrating the possibility of the Hyper-VAM in robotic applications using a single pump.

Closed-loop pneumatic operation can also simplify the control system of the actuator. Instead of using separate regulators for both the positive and negative pressure chambers and controlling

them independently, it is possible to control the Hyper-VAM in closed-loop pneumatic operation using a single pneumatic regulator (Figure 7a). This regulator switches between and controls the airflow being pumped from the negative pressure chamber to the positive pressure chamber and the airflow which is allowed to flow back from the positive pressure chamber to the negative pressure chamber using the atmosphere inlet (ATM) of the vacuum pressure regulator connected to the positive pressure chamber. Using a magnetic linear encoder and implementing proportional integral derivative (PID) control for feedback control of the actuator, the actuator was made to follow a desired trajectory with a payload of 10 kg in closed-loop pneumatic operation without exchanging air with the environment (Figure 7b, Movie S5, Supporting Information).

4.2. Implementation in a Robotic Arm

The Hyper-VAM using closed-loop pneumatic actuation was then implemented as the bicep of a robotic arm with a tendon and pulley system (Figure 7c). A payload of 2 kg at a moment arm of 30 cm was installed onto the arm and closed-loop control was implemented using an angular encoder implemented at the joint

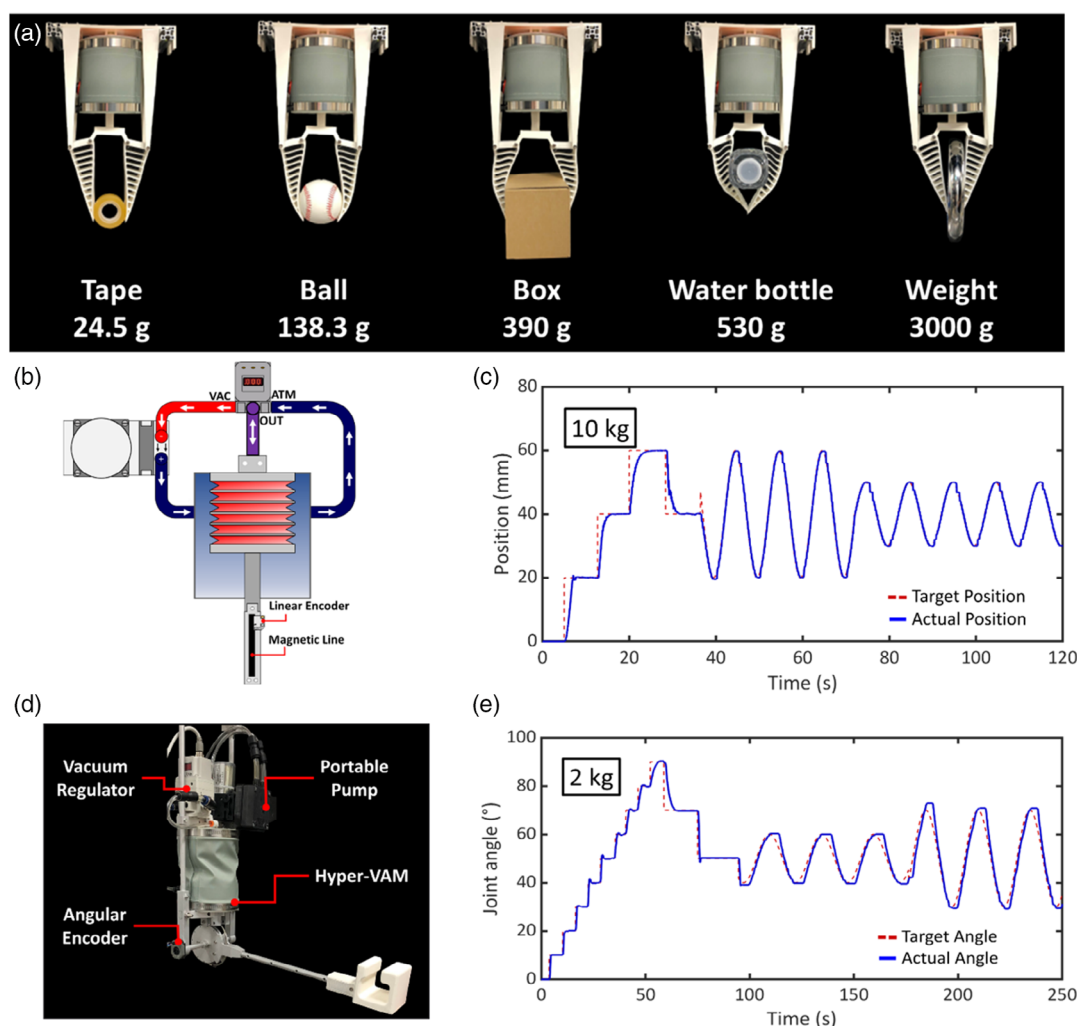


Figure 7. Closed-loop pneumatic operation of the Hyper-VAM. a) Hyper-VAM used to drive a flexible gripper in closed-loop pneumatic operation. b) Setup for the closed-loop pneumatic linear control of the Hyper-VAM. c) Linear control results of the Hyper-VAM in closed-loop pneumatic operation. d) Setup for the robotic arm using closed-loop pneumatic control of the Hyper-VAM. e) Joint control results of the robotic arm using the Hyper-VAM in closed-loop pneumatic operation.

for angular position feedback. The system was able to follow different signals and demonstrates the possibility of using the Hyper-VAM using a single pump and a single regulator for signal tracking in robotic applications (Figure 7d, Movie S6, Supporting Information).

It is to be noted that using a portable pump allows for smaller pressures than using external pumps and that there is a limitation in the amount of air available to be transferred between chambers when in closed-loop pneumatic actuation such that the maximum payload must be subsequently adjusted.

5. Discussion

The Hyper-VAM could lift payloads up to 80 kg with its maximum contraction ratio of 80.2% of the active length, could produce up to $674.93 \text{ J} \cdot \text{m}^{-1}$ of normalized average work per active length, and 101.3 W of mechanical power with a weight of 150 g,

which corresponds to a specific power of 675.3 W kg^{-1} . The construction of the actuator is in function of its intended 80 kg maximum payload which required all parts to be reinforced. One of the main advantages of the actuator for producing high-power actuation is that the actuator's force scales with the cross-sectional area of the negative pressure chamber, unlike many other actuators which require a bundle of similar actuators to increase the force. This allows the actuator to maintain two inlet tubes into the actuator even at scale and to use large inlets to obtain high flow rates. A comparison of the Hyper-VAM with other PAMs is included in Table S1, Supporting Information.

Another significant advantage of the actuator is that the pressure differential is achieved between the positive pressure chamber and the negative pressure chamber unlike other pneumatic actuators where the pressure differential used for actuation is achieved between the inside of the actuator and the atmosphere. This would mean that those actuators would behave differently whether it is operating at high or low altitudes, that they could become hard

to pressurize in highly pressurized environments, or that vacuum-based actuators become entirely unusable in vacuum environments such as in space. The proposed Hyper-VAM can be operated in a vacuum environment without being significantly affected by the external pressure (Figure S1, Supporting Information).

Using a portable pump, the actuator can be operated as a closed-loop pneumatic system. One of the main advantages of closed-loop pneumatic operation is that no air is exchanged with the environment such that the pneumatic system can be more safely operated in cleanrooms, where air passing through the pump should not be transmitted to the atmosphere, or in dirty environments, which could damage the pneumatic system. Different operating gases could also be used to improve the performance of the system and decrease the friction of the gas within the tubes.

6. Conclusion

The Hyper-VAM introduced in this work produces rapid actuation with large forces and high contraction ratios while also being drivable using portable pneumatic pumps. It does not need to be bundled to produce larger forces and uses relatively low pressures. Also, the pressure differential within the actuator is independent of the atmospheric pressure and is thus not affected by changes in atmospheric conditions. This type of negative pressure differential-based actuator could even be operated in space or other vacuumed environments. It has significant potential for application in large-scale soft and hybrid robotic applications as it has the performance necessary in terms of contraction ratio, sustained force throughout the motion, and actuation speed for many real-world applications such as patient and elderly care while retaining the natural compliance of air-driven actuators.

Future work will focus on further increasing the possible pressure differential within the actuator to increase its performance, optimizing the pneumatic driving system, implementing the control and driving strategies necessary for bidirectional actuation which was not demonstrated in this paper, and implementing the proposed Hyper-VAM in human-scale robots for patient care.

7. Experimental Section

Fabrication of Actuator: The rigid elements of the negative pressure chamber including the lateral reinforcements and its top and bottom plates were made from 3D-printed polylactic acid (PLA) filament. The wall of the chamber was made from transparent polyvinyl chloride (PVC) film with a thickness of 0.1 mm. The wall was glued to the top and bottom plates using hot melt adhesives and the lateral reinforcements maintained their position using adhesive tape. The top plate had 3D printed air fitting which allowed air exchange through the top of the actuator.

The rigid elements of the soft hyperbaric chamber including the top and bottom plates were made from 3D-printed PLA filament. The wall of the chamber was made from a technical textile consisting of a nylon fabric coated with a thermoplastic polyurethane (TPU) layer which was thermally sealed onto itself to form a tubular shape. A gasket made from molded Ecoflex 00-30 (Smooth-On Inc.) and held in place using a worm-drive tube clamp was used to join and seal the wall of the chamber with the top and bottom plates. The sleeve joining the bottom plate of the negative pressure chamber to the bottom plate of the hyperbaric chamber consisted of a thermally sealed technical textile and was held in place using rubber

elastics. An air fitting was inserted through a hole made in the wall of the chamber.

Actuator Dimensions: Three actuators with different dimensions were used throughout all experiments where the first used a flexible membrane for sealing of the positive chamber, the second used an O-ring, and the third one was a small version of the first one used in the gripper application and in the vacuum condition implementation. The negative pressure chamber of the first two actuators had a total length of 100 mm, the top and bottom plates each had a thickness of 5 mm, and eight equal side lengths of 40 mm, resulting in a radius of 52.3 mm. The hyperbaric chamber using a flexible membrane had a circular cross section with a radius of 55 mm radius, a total length of 145 mm, and the top and bottom plate each had a thickness of 17 mm. The actuator weighed 150 g. The hyperbaric chamber of the actuator using an O-ring had the same cross section, a total length of 115 mm, and the top and bottom plate each had a thickness of 17 mm. The third actuator had a negative pressure chamber with a total length of 50 mm, the top and bottom plates each had a thickness of 5 mm, and eight equal side lengths of 35 mm. The hyperbaric chamber had a circular cross section with a radius of 50 mm radius, a total length of 110 mm, and the top and bottom plate each had a thickness of 17 mm.

Test Setups for Characterization: The blocked force and isobaric force of the actuator were tested using a linear tensile testing machine (ESM750, MARK-10) equipped with a M5-300 force gauge. For measuring the blocked force, the linear tensile testing machine was moved to the desired position and the force measurements were performed. For measuring the isobaric force, the travel speed of the linear tensile testing machine was set at 0.5 mm s^{-1} . The payload experiments were conducted by attaching the payload to the shaft protruding from the bottom pressure chamber. For the previous experiments, the pressure was supplied to the positive pressure chamber using a pneumatic pump (75/250, Bambi Air Ltd.) and controlled using an electropneumatic regulator (ITV1000, SMC) and for the negative pressure chamber using a vacuum pump (SP770EC, Injae Science Co.) and controlled using an electropneumatic regulator (ITV2090, SMC).

The dynamic range experiments and maximum power experiments were conducted using a custom jig assembled using aluminum frames where the displacement of the payload was measured using a magnetic linear encoder (LM10, RLS). A 5/3 air solenoid valve (VUVG, Festo) was used to direct the flow in out of the chambers during the dynamic range experiment, and a vacuum pump (Rocker 410, Injae Science Co.) and a pneumatic pump (75/250, Bambi Air Ltd.) were used to supply vacuum and pneumatic pressure. For the maximum power experiment, a vacuum pump (EVE-TR, Schmalz) and a pneumatic pump (75/250, Bambi Air Ltd.) were connected directly to the actuator with manual valves opened simultaneously.

The closed-loop control test was performed using a diaphragm pump (SP770EC, Injae Science Co.) with an electropneumatic regulator (ITV2090, SMC) connected to each element, as shown in Figure 4d. The robot arm was made from machined aluminum parts and an encoder was added into its joint (AMT10X, CUI INC.).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

pneumatic artificial muscles, positive and negative pressure actuators, soft actuators, soft robotics, vacuum artificial muscles

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